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Allergic Diseases

Dana M. Tofiq
MBChB; FIBMS; FAAAAI
Asst Professor of Allergy & Immunology

Learning Objectives

By the end of this session, students should be able to:

1. Define allergic diseases and understand the basic mechanisms of the immune system involved in allergic reactions.

2. Identify common allergens that can trigger allergic responses, including foods, airborne particles, insect venom, and medications.

3. Describe immunological basis of allergic diseases, including the role of IgE antibodies, mast cells, and the release of histamines in the allergic response.

4. Identify and describe the common symptoms associated with allergic diseases, including respiratory symptoms, skin reactions, gastrointestinal symptoms, and anaphylaxis.

5. Explore various diagnostic methods used to identify and confirm allergic diseases, including skin tests, blood tests, and elimination diets.

6. Discuss the range of treatment options available for allergic diseases, including allergen avoidance, pharmacotherapy (antihistamines, corticosteroids), and immunotherapy.

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type I → IgE mediated

Allergic Disease (Immediate Hypersensitivity)

Normally, host defense can cope with potentially harmful cells and molecules. Under some circumstances a harmless molecule can initiate an immune response that can lead to tissue damage and death. Such exaggerated, inappropriate responses are termed **hypersensitivity reactions** or **allergic disease**.

normally IgE is mostly produced for parasitic infections

In allergic hypersensitivity, the binding of an antigen to specific IgE bound to its high-affinity receptor on a mast cell surface results in massive and rapid cell degranulation and the inflammatory response. The antigens involved are typically inert molecules present in the environment (these are termed **allergens**).

The immediate effects of allergen exposure are often very florid (early phase response). Allergic disorders also have a second phase, occurring a few hours after exposure and lasting up to several days. These 'late phase responses' (LPRs) are mediated by Th2 cells recognizing peptide epitopes of the allergen. Recruitment of eosinophils is often a prominent feature.

From a pathological and therapeutic viewpoint, the LPR gives rise to chronic inflammation, which is difficult to control. In asthma, the LPR causes prolonged wheezing, which can be fatal. Immediate hypersensitivity is usually responsive to antihistamines but the LPR is not, requiring powerful immune modulators such as corticosteroids.

In immunopathological terms, in the LPR:

1. Neutrophils and eosinophils are prominent in the first 6–18 hours and may persist for 2–3 days.

2. Th2 cells accumulate around small blood vessels and persist for 1–2 days.

3. Mediators responsible for the cellular infiltrate include platelet-activating factor and leukotrienes.

4. Th2 cytokines IL-4 and IL-5 and chemokines such as eotaxin act as growth and activation stimuli for eosinophils, which are capable of extensive tissue damage. Th2 cytokines are also responsible for the class switch of Ig production towards IgE, maintaining the cycle of immediate and late responses.

What makes an allergen so powerful?

Several allergens are proteolytic enzymes, allowing them to cross skin and mucosal barriers. They are often contained within small, aerodynamic particles (e.g. pollen grains) that gain access to nasal and bronchial mucosa.

Why do some people react and others do not?

The tendency to develop allergic responses (known as **atopy**) shows strong heritability. Between 20% and 30% of the UK population is atopic, and two, one or no atopic parents pass on the atopic trait to their children with a risk of 75%, 50% and 15%, respectively.

In developing nations, the tendency to allergy is estimated at one-tenth of the rate in industrialized countries. Among the predisposing genes are those encoding the β chain of the high-affinity receptor for IgE and IL-4, both strongly associated with Th2 pathways. The presence of Th2 cells recognizing allergens is the pathological hallmark of allergy.

What environmental factors are involved?

this idea has lead to trials by injecting parasites to produce immunological maturation

Early exposure to allergens (even in utero) may be a factor in developing atopy. Overzealous attention to cleanliness (the hygiene hypothesis) in developed societies (use of antibiotics, reduced exposure to pathogens that might favor a Th1-like environment) may favor a reduction in Treg activity. This environmental factor is demonstrated by the rapid increase of allergy in the eastern part of Germany following reunification in 1990.

In clinical terms, approximately two-thirds of atopic individuals (who can be identified as those with circulating allergen-specific IgE) have clinical allergic disease (equating to 15–20% of the UK population).

Allergy accounts for up to one-third of school absences because of chronic illness. Allergic disorders include allergic rhinitis (hay fever), allergic eczema, bee and wasp venom allergy, some forms of food allergy, urticaria and angioedema.

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Mediators Involved in the Allergic Response

was already present just liberated

1. Pre-formed mediators → most cells are packed with them

Histamine and serotonin: bronchoconstriction and increased vascular permeability

Neutrophil chemotactic factor (NCF) and eosinophil chemotactic factor (ECF): induction of inflammatory cell infiltration

2. Newly formed mediators (membrane-derived) → after exposure start to form

Leukotriene (LT) B4: chemoattractant

LTC4, LTD4, LTE4 (slow-reacting substance of anaphylaxis, SRS-A): sustained bronchoconstriction and edema

Prostaglandins and thromboxanes: platelet-activating factor (PAF) and prolonged airway hyperactivity

Diagnosis

Diagnosis of allergic disease is usually made on the history and backed up by skin-prick testing (insertion of a tiny quantity of allergen under the skin and measurement of the size of the weal) and/or measurement of serum allergen-specific IgE.

Mast cell tryptase serum levels peak 1–2 hours after an event, remaining high for 24 hours.

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Skin Prick Test

have +ve control → histamine → there must be reaction

have -ve control → normal saline → there must no reaction

Treatment of Allergic Diseases

Avoidance is the first line of therapy. Other measures are as follows:

1. Antihistamines are effective for many immediate hypersensitivity reactions (but have no role in the treatment of asthma).

2. Corticosteroids have several well-identified modifying actions in the allergic process: production of prostaglandin and leukotriene mediators is suppressed, inflammatory cell recruitment and migration are inhibited, and vasoconstriction leads to reduced cell and fluid leakage from the vasculature.

3. Cysteinyl leukotriene receptor antagonists (LTRAs) inhibit leukotrienes (LTs) by blocking the type I receptor (e.g. montelukast, used in asthma, particularly the aspirin-induced type).

4. Omalizumab is a monoclonal antibody that binds IgE. It is used in severe asthma that cannot be controlled with a corticosteroid plus a long-acting β2 agonist. Treatment must be initiated in a specialist center where staff have experience of treating severe, persistent asthma.

5. Dupilumab is a monoclonal antibody that inhibits the biological activity of both IL-4 and IL-13. It has been shown to be beneficial in severe atopic eczema and asthma.

6. Desensitization (allergen immunotherapy) works on the principle that allergy can be prevented by inoculation if the allergen is given in a controlled way. Desensitization is indicated for disorders in which the hypersensitivity is IgE-mediated, e.g. life-threatening allergy to insect stings, drug allergy and allergic rhinitis. An induction course of subcutaneous injections of increasing doses of the allergen extract, given once every 1–2 weeks, is followed by maintenance injections monthly for 2–3 years.

From an immunological viewpoint, desensitization seems capable of modifying the allergic response at several levels. Sublingual allergen immunotherapy (using grass pollen extract tablets of Phl p 5 from timothy grass) is used in hay fever that has not responded to anti-allergic drugs (the first dose is given under medical supervision).

Mechanism of Action of Desensitization for Allergy

Probable mechanism	Effect
IgG blocking antibodies	During repeated exposure to desensitizing allergen, IgG class antibodies develop (especially IgG4 subclass); these compete with the pathogenic IgE for allergen binding and/or prevent IgE–allergen complexes from binding to mast cell high-affinity IgE receptors
Regulation	Exposure to repeated desensitizing allergen induces Treg cells, which recognize allergen but invoke regulatory immune responses, dampening down migration, infiltration and inflammation
Immune deviation	A shift away from Th2- to Th1-producing CD4 cells results in the generation of cytokines (e.g. interferon-gamma) that are inhibitory to IgE production

Anaphylaxis

Anaphylaxis is a serious allergic reaction that is rapid in onset and may cause death. It arises as an acute, generalized IgE-mediated immune reaction involving specific antigen, mast cells and basophils. The reaction requires priming by the allergen, followed by re-exposure. To provoke anaphylaxis, the allergen must be systemically absorbed, after either ingestion or parenteral injection. A range of allergens that provoke anaphylaxis has been identified.

Anaphylaxis is rare, but the symptom/sign constellation ranges from widespread urticaria to cardiovascular collapse, laryngeal oedema, airway obstruction and respiratory arrest leading to death.

Fatal reactions to penicillin occur once every 7.5million injections.

Between 1 in 250 and 1 in 125 individuals have severe reactions to bee and wasp stings, and a death takes place every 6.5 million stings, with 60–80 deaths per year in North America and 5–10 in the UK.

Sources of Allergens Known to Provoke Anaphylaxis

Foods

- Nuts: peanuts (protein – *Arachis hypogaea* Ara h 2), Brazil, cashew
- Shellfish: shrimp (allergen Met e 1), lobster
- Dairy products
- Egg
- More rarely: citrus fruits, mango, strawberry, tomato

Venoms

- Wasps, bees, yellow-jackets, hornets

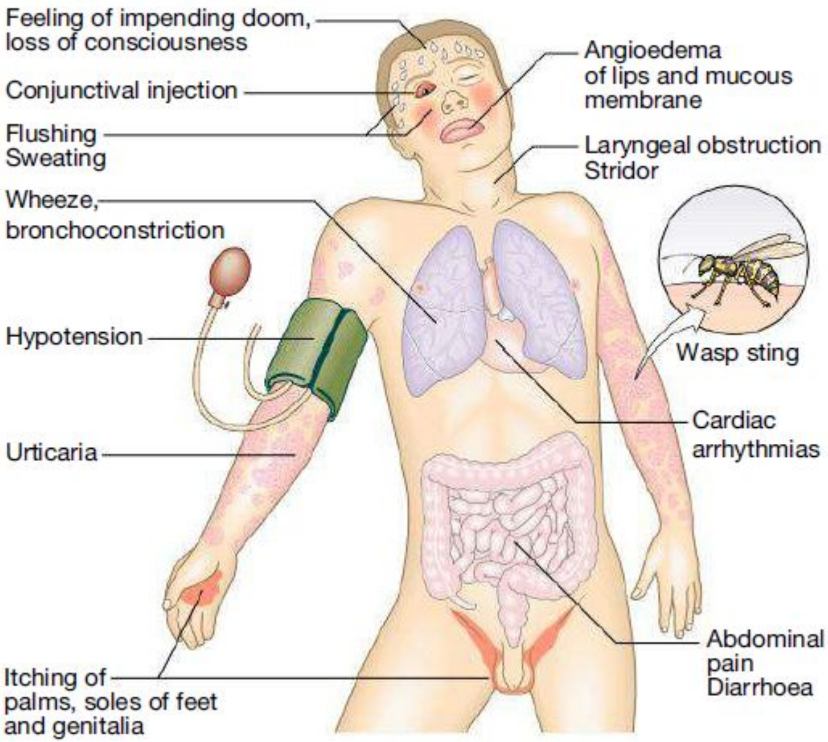
Medications

- Antisera (tetanus, diphtheria), dextran, latex, some antibiotics

Central to the pathogenesis of anaphylaxis is the activation of mast cells and basophils, with systemic release of some mediators and generation of others. The initial symptoms may appear innocuous: tingling, warmth and itchiness. The ensuing effects on the vasculature lead to vasodilation and edema. The consequence may be no more than a generalized flush, with urticaria and angioedema. More serious sequelae are hypotension, bronchospasm, laryngeal edema and cardiac arrhythmia or infarction. Death may occur within minutes.

Serum platelet-activating factor (PAF) levels correlate directly with the severity of anaphylaxis, whereas PAF acetylhydrolase (the enzyme that inactivates PAF) correlates inversely and is significantly lower in peanut-sensitive patients with fatal anaphylactic reactions.

Clinical Manifestations of Anaphylaxis



Management

Early recognition and treatment are essential. The best treatment is prevention. Avoidance of triggering foods, particularly nuts and shellfish, may require almost obsessive self-discipline. Patient education is necessary and many are instructed in the self-administration of adrenaline (epinephrine) and carry preloaded syringes. Desensitization has a well-established place in the management of this disorder, particularly if exposure is unavoidable or unpredictable, as in insect stings.

In peanut allergy, it has been shown that early introduction of peanuts in children modulates the response in later life, and this benefit extends to high-risk children with atopic eczema or allergy to another food. Further studies have shown that it is possible to give graded peanut exposure to those already sensitized and that oral desensitization can reduce the risk of anaphylaxis associated with inadvertent exposure in children with a history of peanut allergy. The addition of modified peanut peptides further enhances the success and tolerability of this approach.

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Treatment of Acute Anaphylaxis

Clinical features

- Bronchospasm
- Facial and laryngeal oedema
- Hypotension
- Nausea, vomiting and diarrhoea

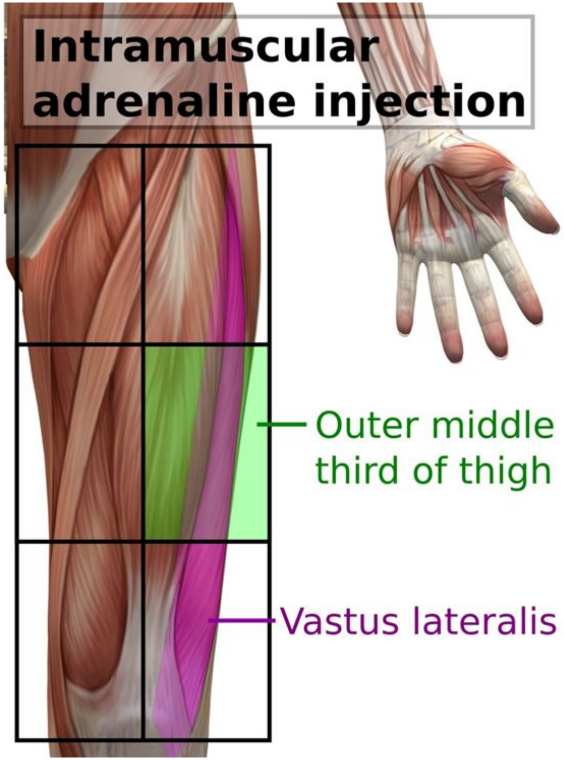
Management

- ABCDE (airway, breathing, circulation, disability, exposure)
- Position the patient lying flat with the feet raised
- Ensure the airway is free
- Give oxygen
- Monitor blood pressure
- Establish venous access
- Administer 0.5 mg *intramuscular* adrenaline (epinephrine) 0.5 mL of 1 : 1000 adrenaline, i.e. 1 mg/mL, and repeat after 5 min if shock persists (dose reduced for children or adults on beta-blockers)
- Administer intravenous antihistamine (e.g. 10–20 mg chlorphenamine) slowly
- Administer 100 mg intravenous hydrocortisone.
  - If hypotension persists, give 1–2 L of intravenous fluid
  - If hypoxia is severe, assisted ventilation may be required
  - Take blood for tryptase levels (aids diagnosis)

EpiPen® Auto-Injector 0.3 mg

EpiPen® Auto-injector 0.3 mg

Pediatric dose is half dose of adults



How to prescribe self-injectable adrenaline (epinephrine)

Prescription (normally initiated by an immunologist or allergist)

- Specify the brand of autoinjector, as they have different triggering mechanisms
- Prescribe two devices

Indications

- Anaphylaxis to allergens that are difficult to avoid:
  - Insect venom
  - Foods
- Idiopathic anaphylactic reactions
- History of severe localised reactions with high risk of future anaphylaxis:
  - Reaction to trace allergen
  - Likely repeated exposure to allergen
- History of severe localised reactions with high risk of adverse outcome:
  - Poorly controlled asthma
  - Lack of access to emergency care

Patient and family education

- Know when and how to use the device
- Carry the device at all times
- Seek medical assistance immediately after use
- Wear an alert bracelet or necklace
- Include the school in education for young patients

Other considerations

- Caution with β-adrenoceptor antagonists (β-blockers) in anaphylactic patients as they may increase the severity of an anaphylactic reaction and reduce the response to adrenaline (epinephrine)

Reference:

Kumar and Clark's Clinical Medicine, 10th Edition
Authors : Adam Feather & David Randall & Mona Waterhouse
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